

Team Control Number

14255

Problem Chosen

C

1. Summary

The upcoming 2024 Paris Summer Olympics will see more spectators than ever. The IOC (International Olympics Committee) has tasked our team at the MidMCM Olympic Consultants to create a model for the ceremony schedule and to find out how many medals and flags they will need to avoid public embarrassment, while also having a reasonable budget. With not much to go with, many assumptions have been made. These include ceremony timings, strange exceptions, etc. We have also organized the schedule into 5-minute blocks. The reason for this is that they would make our schedule more concise and easier to read.

We have been asked to choose a venue to get started with, the La Defense Arena, since swimming and water polo have basic solutions for medals and flags. To calculate the number of flags we needed, we had to start with some basic reasoning. 203 countries are participating in the Olympics, and getting flags for all of them seems too expensive since we know a flag can cost at least \$900. Based on this, we then collected data from the past 4 data sets of Olympic swimming and water polo. This data was based on which countries got medals each year and how many they got. Every country that has placed on the podium will get 3 flags and every country that has placed 4th will get 1 flag in case they manage to win. Altogether, there will be 105 flags for 39 countries. We also took a look at ties to make sure we were not short of any medals. To calculate how many medals we would need, normal distribution and standard deviation were used, to find the chance of ties and to calculate a set of medals needed using:

$$\sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}, \bar{x} = \frac{\sum_{i=1}^n x_i}{n}.$$

This method of calculation allows for 95% certainty when predicting the number of medals in order to account for ties. This means our final numbers will be as accurate as possible. These calculations came out with 26 sets of medals for 16 events. All of our data was organized into data tables to provide legibility and easy understanding.

Our model and solution should be easily applied to the other events and arenas as well. If data and calculations are done similarly, flags and medals can be found, and the only change would be the timings on the schedule. This flexibility along with the simple concepts used are among the strengths. The only weakness we have is that there were no clear starting times or playing brackets for us to work off of, so there may be some changes when actually being used.



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2. SCHEDULE

DAY 1 SWIMMING

TIME INTERVAL (starts at 10 but have a 5-minute late start in case of any problems)	EVENT / AWARD CEREMONY
10:05 - 10:07	100 butterfly
3-minute break	
10:10 - 10:13	200 butterfly
2-minute break	
10:15 - 10:30	Award Ceremony for 100 butterfly
5-minute break	
10:35 - 10:50	Award Ceremony for 200 butterfly
5-minute break	
10:55 - 10:57	100 Backstroke
3-minute break	
11:00 - 11:03	200 Backstroke
2-minute break	
11:05 - 11:20	Award Ceremony for 100 Backstroke
5-minute break	
11:25 - 11:40	Award Ceremony for 200 Backstroke
5-minute break	
11:45 - 11:47	100 Breaststroke
3-minute break	
11:50 - 11:53	200 Breaststroke
2-minute break	
11:55 - 12:10	Award Ceremony for 100 Breaststroke
5-minute break	



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12:15- 12:30	Award Ceremony for 200 Breastroke
5- minute break	
12:35- 12:37	100 freestyle
3- minute break	
12:40- 12:43	200 freestyle
2- minute break	
12:45- 1:00	Award Ceremony for 100 Freestyle
5- minute break	
1:05- 1:20	Award Ceremony for 200 Freestyle

SCHEDULE CONT.

DAY 1 WATER POLO

2:00- 2:08	1st period (Country A vs Country B)
2- minute break	
2:10- 2:18	2nd period (Country A vs Country B)
2- minute break	
2:20- 2:28	3rd period (Country A vs Country B)
2- minute break	
2:30- 2:38	4th period (Country A vs Country B)
2- minute break	
2:40- 2:55	Award Ceremony for Country A vs Country B
5 minute break	
3:00- 3:08	1st period (Country C vs Country D)
2- minute break	
3:10- 3:18	2nd period (Country C vs Country D)
2- minute break	



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3:20 - 3:28	3rd period (Country C vs Country D)
2 - minute break	
3:30 - 3:38	4th period (Country C vs Country D)
2 minute break	
3:40 - 3:55	Award Ceremony for Country C vs Country D
2 - minute break	Winner of Country C vs D moves on to next round
3:40 - 3:48	1st period (winners vs winners)
2 - minute break	
3:50 - 3:58	2nd period (winners vs winners)
2 - minute break	
4:00 - 4:08	3rd period (winners vs winners)
2 - minute break	
4:10 - 4:18	4th period (winners vs winners)

* whoever wins the match between A, B, and will move on to face the winner between Country C and D, and whoever wins that match will get the medal. This will be the same schedule for water polo that will happen every day since we do not know the specific country.

SCHEDULE DAY 2

TIME INTERVAL	EVENT / AWARD CEREMONY
10:05 - 10:07	50m freestyle
3 - minute break	
10:10 - 10:15	400 m freestyle
5 - minute break	
10:20 - 10:30	800m freestyle
5 - minute break	



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10:30 - 10:50	1500 m freestyle
5-minute break	
10:55 - 11:10	Award Ceremony for 50 m free
5-minute break	
11:15 - 11:30	Award Ceremony for 400 m free
5-minute break	
11:35 - 11:50	Award Ceremony for 800 m free
5-minute break	
11:55 - 12:10	Award Ceremony for 1500 m free
5-minute break	
12:15 - 12:18	200 m individual medley
2-minute break	
12:20 - 12:25	400m individual medley
5-minute break	
12:30 - 12:45	Award Ceremony for 200m medley
5-minute break	
12:50 - 1:05	Award Ceremony for 400m medley
10-minute break * relays start so extra time to prepare	
1:15 - 1:20	4x100m medley relay
5-minute break	
1:25 - 1:30	4x100m Freestyle Relay
5-minute break	
1:35 - 1:40	4x200m Freestyle Relay
5-minute break	
1:45 - 2:00	Award Ceremony for 4x100m medley relay



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5-minute break	
2:05- 2:20	Award Ceremony for 4x100 Freestyle Relay
5-minute break	
2:25- 2:40	Award Ceremony for 4x200m Freestyle relay

SCHEDULE CONT. WATER POLO

DAY 2

For water polo, it would be the same schedule as day one, since we cannot confirm which countries would play against which countries. Because of that, we have a general schedule for how the water polo games should look like.

3. Introduction and Problem Restatement

Whether you're a sports fan or not, you've probably heard of the Olympics. The Olympics is one of the most popular events ever, with a global audience of over 3 billion people. 2024 brings in another Summer Olympics to be hosted in Paris. 2016 saw 70,000 spectators and 2020 saw 18,000 spectators. However, 2024 is expected to have 600,000 spectators and a drastic increase in the total audience. The increase can also be credited to the growing popularity of the Olympics.

However, the Olympic games and their committee have received backlash for their major success. As more and more countries participate in the Olympics, affording medals, flags, and overall materials may become complicated and expensive. It becomes impractical to have a flag for every participating country and to have an excessive amount of medals, especially in economies of large countries where minimizing costs is a primary goal. In addition, with gold medals costing up to \$750 and silver medals costing up to \$75 in the economy, funding just one event with 3 winners would cost almost \$1000. Added alongside team events, the number of events overall, and the possibility of there being more than three winners, funding even just swimming and water polo, the two sports at our chosen venue, would cost incomprehensible amounts.

To save money, the IOC has asked us at MidMCM to help them develop a schedule and determine the flags and medals needed for the 2024 Paris Olympics to maximize cost-efficiency.

- Schedule: This part will contain the schedule we will provide to the IOC.
- Data Organization: This section will contain several charts organizing our data, including our calculations and probabilities for each country attaining a place on the podium. It will include a detailed explanation on the number of medals (gold, silver, and bronze) required.



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- Letter: The final component will be the letter we send to the IOC. There will be brief highlights and a general explanation on the specifics of our decisions, as well as an explanation of our

4. Assumptions and Justifications

1. Attendance

Assumption: We assume every invited country will show up.

Justification: The Olympics are an incredibly important event for the participants as well as the country as they bring glory and pride. No invited contestants will wish to skip.

2. No Emergencies

Assumption: We assume that no medical emergencies, such as drowning, will occur.

Justification: The Olympics consist of the best athletes in the world. It seems highly improbable that one of them would have health issues or unforeseen injuries.

3. Number of Team Members

Assumption: We assume that water polo teams will each have 13 players.

Justification: Based on research, water polo teams will have either 7 players in the water at one time and up to 6 reserves, or 12 players and up to 1 AP alternative athlete. The number of medals needed will depend on how many team members there are, so it will be best to have a safe and standardized number for the number of team members; additionally, having more players to choose from is always beneficial.

4. Swimming Events

Assumption: We assume that the events for the swimming portion will be as follows:

- 50m Freestyle (women's / men's)
- 100m Freestyle (women's / men's)
- 200m Freestyle (women's / men's)
- 400m Freestyle (women's / men's)
- 800m Freestyle (women's / men's)
- 1500m Freestyle (women's / men's)
- 100m Backstroke (women's / men's)
- 200m Backstroke (women's / men's)
- 100m Breaststroke (women's / men's)
- 200m Breaststroke (women's / men's)
- 100m Butterfly (women's / men's)
- 200m Butterfly (women's / men's)
- 200m Individual Medley (women's / men's)



- 400m Individual Medley (women's / men's)
- 4x100m Freestyle Relay (women's / men's)
- 4x200m Freestyle Relay (women's / men's)
- 4x100m Medley Relay (women's / men's / mixed)

Justification: These were found through research. All of these events appeared in the previous Summer Olympics, except for 1500m women, 800m men, and 4x100m mixed medley. However, these three events were played in the 2020 Tokyo Olympics and will most likely appear in the 2024 Olympics.

Addendum: We assume that events occur in IM order; that is, Butterfly, Backstroke, Breaststroke, and Freestyle.

5. Timings

Assumption: We assume that our schedule might take more than one day, and we assume that we will take about 2-4 days to complete all games and award ceremonies.

Justification: Usually, in the past 4 years of Olympic games, the games have taken around a few days, in order to complete all the swimming events. Since we assumed quite a few different events will occur, they must all take place on different days.

According to the recently published Paris 2024 schedule, they have planned for 11 days of different swimming events.

6. No Strange Exceptions

Assumption: We assume that there will not be a case where a country that has not placed a podium in previous years suddenly places podium.

Justification: Such a case is extremely unlikely, given that the country has not regularly placed podium in the past 4 Olympic games.

7. Reusing Flags

Assumption: We assume that the contestants cannot keep the flags, and therefore, the flags can be reused for other events.

Justification: It would cost extra money to buy more flags, so it would be more convenient and reasonable to reuse flags in order to minimize spending money.

8. Ceremony Timing

Assumption: We assume that the award ceremony for an event will happen directly after the finals of the event finish. We are also organizing ceremonies after a group of similar events. For example, we would finish the 100 fly and 200 fly, then do the ceremonies for both. Relays and medleys will take place at the end. Each ceremony will take 15 minutes to ensure full protocol, allow for reporters, photoshoots, and brief interviews, and to make sure no mishaps can occur.

Justification: This is the way it worked in the past.



9. Start of events

Assumption: We assume that the swimming events would start around 10 AM, so that there would be enough time to complete a reasonable amount of events each day, including both swimming and water polo.

Justification: This is a similar time schedule that has been used in the past years.

10. Breaks and Time Increments

Assumption: We assume that there will be breaks between each swimming event. Each break will be a certain but variable length so that the schedule can be easily organized into 2-15 minute blocks. We will have a 30-minute gap between the last swimming event of the day and the first water polo event. There will also be a break between changes of events in terms of stroke. For example, when we finish the last butterfly events, we will take a break and then continue on to backstroke.

Justification: This way the athletes get some rest, and it also makes the schedule easier to read and implement.

11. Pools

Assumption: There will be one pool at the La Défense Arena.

Justification: The layout of the area provides one pool in the center and seats around it. This was found out through research and finding images online.

12. Time Intervals

Assumption: We assume that the maximum time for the 100m would be 2 minutes. 200m would be around 3 minutes and the 4x100m will take around 5 minutes. We also assume that the maximum time for the 800 and 1500 would be around 10-20 minutes. Justification: An Olympic swimmer should be able to complete their events under these time intervals but just in case of any difficulties or a slow swimmer, we have given them the maximum amount of time they would need to finish the events,

13. Water Polo Games

Assumption: We assume the water polo games would have 4 games of 8 minutes each for men and women.

Justification: These are the rules that FINA has set out for water polo, it is also a Reasonable time for each game.

14. Change of Sports

Assumption: When there is a change of sports we would have a 40-minute gap, so when we change from swimming to water polo there is a 40-minute gap.



Justification: This would be a way to make sure we have no mishaps and so we can smoothly change between events, this is also a similar thing that has happened in past Olympic games.

15. Overtime (for water polo)

Assumption: We assume that there will be a 2-minute limit to the overtime in water polo, overtime happens when countries tie after the 4 periods and have to have a tiebreaker.

Justification: This is how FINA set it out in their rulebooks, and 2 minutes also seems like a fair enough time to complete a tiebreaker.

16. Specific Countries

Assumption: Since water polo is a game where two countries are forced to face against each other we need to know the specific countries that are playing against each other. Unfortunately, we cannot specifically say what countries will take part against each other.

Justification: The schedule for what countries taking place in Paris 2024 has not been released yet, so we can't know who to place against whom. It would also be completely arbitrary to place a random country against a random country because according to past games, the Olympics places countries against each other based on their similarities in skill.

17. Same Schedule for Both Men's and Women's Events

Assumption: We assume that the schedule would be the same for both men and women.

Justification: We know the events are the same so logically, there should be no change for events between men and women.

5. Variables Definition

Variable	Definition
x_i	The i th value in set $\{x\}$
$\bar{x}$	The mean of set $\{x\}$



6. Presentation of Model and Solution

There will be 3 flags for every country that has placed podium at least once in the past 4 Summer Olympic Games. This is because they have the highest chance of placing on the podium again. There will be 1 flag provided for other countries that have placed 4th (and not on the podium) at least once in the past 4 summer Olympic games, because those countries all have a reasonable chance of placing on the podium in the 2024 games. This method minimizes the amount of required flags, while still relatively guaranteeing the supply of all flags required. The flags required are as follows:

- Australia (3)
- Austria (3)
- Belarus (3)
- Belgium (3)
- Brazil (3)
- Canada (3)
- China (3)
- Denmark (3)
- Finland (3)
- France (3)
- Germany (3)
- Great Britain (3)
- Hong Kong (3)
- Hungary (3)
- Italy (3)
- Japan (3)
- Kazakhstan (3)
- Lithuania (3)
- Netherlands (3)
- Norway (3)
- ROC (3)
- Serbia (3)
- Singapore (3)
- Slovenia (3)
- South Africa (3)
- South Korea (3)
- Spain (3)
- Sweden (3)
- Switzerland (3)
- Tunisia (3)
- Ukraine (3)
- USA (3)
- Zimbabwe (3)
- Croatia (1)



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- Greece (1)
- Montenegro (1)
- New Zealand (1)
- Poland (1)
- Romania (1)

Data tables (data collection research): this data was collected to see how many medals each country got based on events. If a country got no medals whatsoever, we would eliminate them, with the exception of the countries who came in 4th place, because they do have a reasonable chance of placing on the podium.

A yellow background indicates gold, silver background indicates silver, orange background indicates bronze, and a blue background indicates 4th.

WIP Women				2008				2012				2016				2020			
				gold	silver	bronze	4th	gold	silver	bronze	4th	gold	silver	bronze	4th	gold	silver	bronze	4th
Netherlands				1															
USA					1			1				1				1			
Australia						1				1									
Spain									1								1		
Italy													1						
Russia														1					
Hungary							1				1				1			1	1

WIP Men				2008				2012				2016				2020			
				gold	silver	bronze	4th	gold	silver	bronze	4th	gold	silver	bronze	4th	gold	silver	bronze	4th
USA						1													1
Spain																			
Italy									1					1					
Hungary				1														1	
Serbia						1				1		1				1			
Greece								1									1		
Croatia													1						
Montenegro							1				1				1				

women's				2008				2012				2016				2020			
				gold	silver	bronze	4th	gold	silver	bronze	4th	gold	silver	bronze	4th	gold	silver	bronze	4th
Australia				12	1	7		4		10	2	2	4			1		1	1
Canada						1						1	1						
USA				2	13	9		3	14	3	6	2	14	7		4		2	2
Hungary												1	3			1		1	1
China				1	5	5		2	3	1	3							2	4
Japan						1				1	7							1	
Zimbabwe				1	3								1						
Austria						1		1											
Germany				2										1					
Norway						1													
Slovenia					1														
Italy				1	1			1											1
Great Britain				2		1		2		2		1			3				3
Netherlands				4				1	4	1									1
Denmark						1						2	1					4	
Lithuania								1											
Belarus									2									1	
ROC								3		1		2		2		2			1
Spain									2				2						1
France								5	1				1					3	
Sweden															1				
South Africa																			
Hong Kong																			
Finland								1											
Poland								1											
Romania								1											
Jamaica												1							
New Zealand												1							

mixed				4x100m medley relay (2020)			
United Kingdom				1			
China					1		
Australia						1	
Italy							1



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2020				800m freestyle			
gold	silver	bronze	4th	gold	silver	bronze	4th
2	2	10	2				
			1				
13	2	1	2	1			
1	1		2				
1							
	1		1				
			1				
		1	1				1
	4	6	1		1		
6	6	1	2				
	1	1					
1	4	1	3				
	2		1				
		1					
			1				
		2					
1							
		2					
	1					1	
25	24	26	AUT, ROU	1	1	1	

To determine how many medals will be needed, we will use normal distribution and standard deviation to find the optimal number of medals so that, with 95% confidence, we can buy just enough for swimming events, including the possibility of ties. Results were taken from the previous 7 Olympics to determine how often ties occurred. The below tables show the medals awarded for each event in 2020, 2016, 2012, 2008, 2004, 2000, and 1996.

Note that the number within the grid indicates the *place*, not the amount.



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	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	2016																	
2	men																	
3	100m back	100m breast	100m fly	100m free	200m back	200m breast	200m fly	200m free	200m IM	400m free	400m IM	4x100m free	4x100m IM	4x200m free	50m free	1500m free		
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
6	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
7																		
8	women																	
9	100m back	100m breast	100m fly	100m free	200m back	200m breast	200m fly	200m free	200m IM	400m free	400m IM	4x100m free	4x100m IM	4x200m free	50m free	800m free		
10	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
11	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
12	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
13																		
14	2020																	
15	men																	
16	100m back	100m breast	100m fly	100m free	200m back	200m breast	200m fly	200m free	200m IM	400m free	400m IM	4x100m free	4x100m IM	4x200m free	50m free	1500m free	800m free	
17	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
18	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
19	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3 mixed
20	women																	
21	100m back	100m breast	100m fly	100m free	200m back	200m breast	200m fly	200m free	200m IM	400m free	400m IM	4x100m free	4x100m IM	4x200m free	50m free	800m free	1500m free	
22	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
23	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
24	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
25																		
26	2012																	
27	men																	
28	100m back	100m breast	100m fly	100m free	200m back	200m breast	200m fly	200m free	200m IM	400m free	400m IM	4x100m free	4x100m IM	4x200m free	50m free	1500m free		
29	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
30	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
31	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
32	women																	
33	100m back	100m breast	100m fly	100m free	200m back	200m breast	200m fly	200m free	200m IM	400m free	400m IM	4x100m free	4x100m IM	4x200m free	50m free	800m free		
34	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
35	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
36	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
37																		
38	2008																	
39	men																	
40	100m back	100m breast	100m fly	100m free	200m back	200m breast	200m fly	200m free	200m IM	400m free	400m IM	4x100m free	4x100m IM	4x200m free	50m free	1500m free		
41	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
42	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
43	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
44																		
45	women																	
46	100m back	100m breast	100m fly	100m free	200m back	200m breast	200m fly	200m free	200m IM	400m free	400m IM	4x100m free	4x100m IM	4x200m free	50m free	800m free		
47	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
48	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
49	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
50																		
51	2004																	
52	men																	
53	100m back	100m breast	100m fly	100m free	200m back	200m breast	200m fly	200m free	200m IM	400m free	400m IM	4x100m free	4x100m IM	4x200m free	50m free	1500m free		
54	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
55	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
56	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
57																		
58	women																	
59	100m back	100m breast	100m fly	100m free	200m back	200m breast	200m fly	200m free	200m IM	400m free	400m IM	4x100m free	4x100m IM	4x200m free	50m free	800m free		
60	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
61	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
62	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
63																		
64	2000																	
65	men																	
66	100m back	100m breast	100m fly	100m free	200m back	200m breast	200m fly	200m free	200m IM	400m free	400m IM	4x100m free	4x100m IM	4x200m free	50m free	1500m free		
67	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
68	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
69	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
70																		
71	women																	
72	100m back	100m breast	100m fly	100m free	200m back	200m breast	200m fly	200m free	200m IM	400m free	400m IM	4x100m free	4x100m IM	4x200m free	50m free	800m free		
73	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
74	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
75	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
76																		
77	1996																	
78	men																	
79	100m back	100m breast	100m fly	100m free	200m back	200m breast	200m fly	200m free	200m IM	400m free	400m IM	4x100m free	4x100m IM	4x200m free	50m free	1500m free		
80	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
81	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
82	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
83																		
84	women																	
85	100m back	100m breast	100m fly	100m free	200m back	200m breast	200m fly	200m free	200m IM	400m free	400m IM	4x100m free	4x100m IM	4x200m free	50m free	800m free		
86	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
87	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
88	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3

The number of ties was summed. For the three team events, shown by variable m, there were no ties whatsoever out of 42 samples. For the 13 individual events, shown by variable n, there were 7 ties out of 182 samples. Upon calculating the standard deviation using the formula for standard deviation,

$$\sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}, \text{ where } \bar{x} \text{ is the average of all } x, \text{ i.e., } \frac{\sum_{i=1}^n x_i}{n}.$$



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Standard deviation is the process of measuring how spread out numbers are. In our case, it was a simple solution, because using standard deviation would allow a range of flags and medals, which would be 95% certainly accurate.

All Medals Given Out in Singles Events

Note: the number in parenthesis indicates the amount of ties.

Mens 2016	Womens 2016	Mens 2020	Womens 2020	Mens 2012	Womens 2012	Mens 2008	Womens 2008	Mens 2004	Womens 2004	Mens 2000	Womens 2000	Mens 1996	Womens 1996
123	1233 (1)	123	123	123	123	1233 (1)	123	123	123	123	123	123	123
123	123	123	123	123	123	123	123	123	123	123	123	123	123
1222 (3)	123	123	123	123	123	123	123	123	123	123	123	123	123
123	113 (1)	123	123	123	123	1233 (1)	123	123	123	1233 (1)	123	123	123
123	123	123	123	123	123	123	123	123	1233 (1)	123	123	123	123
123	123	123	123	123	123	123	123	123	123	123	123	123	123
123	123	123	123	123	123	123	123	123	123	123	123	123	123
123	123	123	123	123	123	123	123	123	123	123	123	123	123
123	123	123	123	123	123	123	123	123	123	123	123	123	123
123	123	123	123	123	123	123	123	123	123	123	123	123	123
123	123	123	123	123	123	123	123	123	123	123	123	123	123
123	123	123	123	123	123	123	123	123	123	123	123	123	123
123	123	123	123	123	123	123	123	123	123	123	123	123	123
123	123	123	123	123	123	123	123	123	123	123	123	123	123
123	123	123	123	123	123	123	123	123	123	123	123	123	123
3	2	0	0	0	0	2	0	0	1	1	0	0	0

Among 182 events, 2 of them contained one tie, 2 of them contained two ties, and 1 of them contained three ties.

Thus, we must calculate the standard deviation of

$\{ \frac{0, 0, \dots, 0}{177 \text{ 0's}}, 1, 1, 2, 2, 3 \}$.

This can be done with the simple code

```
def calc_mean(nums):
    total = sum(nums)
    mean = total / len(nums)
    return mean
```

This portion calculates the total using the built-in sum() function and then calculates the mean by dividing the total by the amount of numbers in the set, calculated using the built-in len() function.

```
def calc_variance(nums):
    #recall that variance is just the square of the standard deviation.
    mean = calc_mean(nums) #define mean
    sd = [(x - mean) ** 2 for x in nums] #build a set sd; for every element x in nums,
    subtract it from the mean, and then square the result.
    variance = sum(sd) / len(nums) #divide the sum of the set sd by the number of
    elements in the set nums.
    return variance
```

```
def stdv(nums):
    variance = calc_variance(nums) #define variance
```



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```

standard_deviation = variance ** 0.5 #calculate the square root of variance
return standard_deviation

data_set = [3, 2, 2, 1, 1] #the initial data set contains the five non-zero values
for i in range(173):
    data_set.append(0) #append 177 0's into the data set

std_dev = stdv(data_set) #calculate the standard deviation of the data set
print(std_dev) #output the standard deviation

```

We found the standard deviation of ties in all individual events to be 0.3228, which when added to the mean of the set, 0.0506, was 0.3734. This was rounded up to 1 (since, obviously, we cannot provide 0.3228 medals). In other words, in the worst scenario, it is expected that only a single tie would happen. The standard deviation for the team events turned out to be 0, since not a single tie happened in any team events. This meant that no tie was possible for team events, or, at the very least, no tie was reasonably likely. We used the total number of events for teams and individuals and the standard deviations to calculate how many sets of medals we would need, with one set being one gold, one bronze, and one silver medal. With 13 individual events each requiring one set of medals, and with one tie being expected, 14 sets of medals would be provided. This is because, while we know a tie could happen, we do not know in which place the tie would occur. With 3 team events or relays, each requiring 4 sets of medals for the 4 team members, 12 sets of medals would be provided. Since no tie is expected, no extra medals will be needed. This brings the total number of sets of medals to 26 sets of medals across 16 events. Additionally, to account for the truly unexpected, a surplus of two gold medals, two silver medals, and two bronze medals will also be purchased. In total, we used 28 sets of medals (gold, silver, bronze).

Water polo presents much less of a challenge. Since it is assumed that all countries will bring all 13 players that they *can* bring, it can be assumed that for the three winners, 39 sets of medals are required. Since ties are impossible in water polo, there is no need for a surplus.

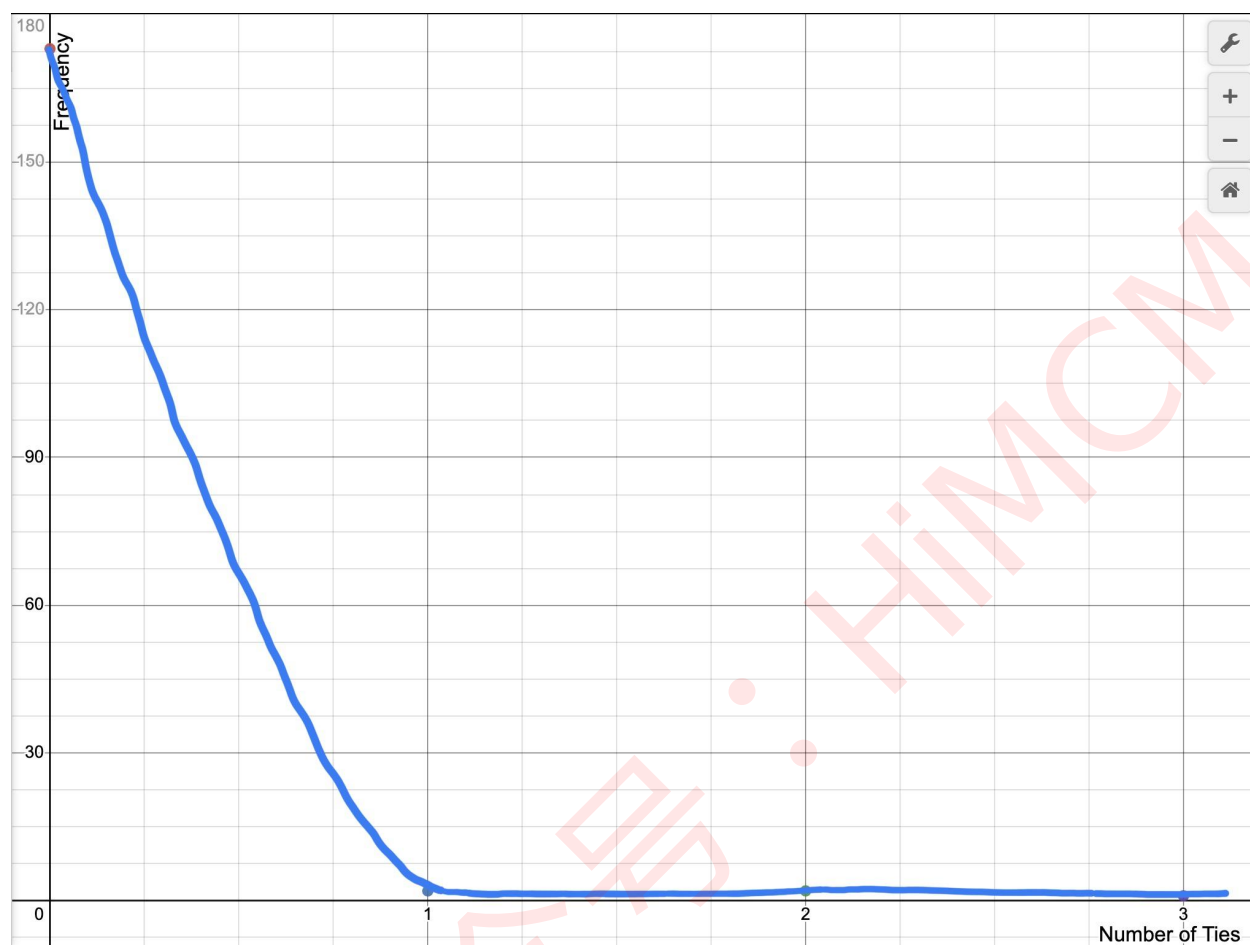
Summing water polo and swimming, we have a final of 67 sets of medals, meaning 67 gold medals, 67 silver medals, and 67 bronze medals. This is the *optimal* amount.

At minimum, we can assume no ties and no required surplus. Therefore, we would have a minimum of 25 sets + 39 sets = 64 sets of medals. This is the *minimal* amount.

At maximum, we can assume every event that can tie ties. Therefore, we would have 2x13 sets + 12 sets + 39 sets = 77 sets of medals. This is the *maximum* amount.



Number of Ties vs Frequency



7. Analysis of Work

7.1 Reflection

A similar model of the schedule would work for the other two arenas. The breaks would be useful when applied to the other sports as well. The only change that would happen would be the amount of time required for each event. The format would be mostly the same because both arenas consist of one sport that has many events and another one that only has one event for men and another for women.

If data was collected and analyzed similarly for the other events, medals, and flags could also be predicted in the exact same way.



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7.2 Strengths

Because we used accurate data and statistics, our solution would have a high accuracy rate. The concepts we used are also not too complex, so it's unlikely for there to be unseen mathematical errors.

Our solution can also easily be adapted and applied for other sports and arenas.

The schedule is also easy to follow and flexible in case any modifications are needed.

7.3 Weaknesses

We also have some weaknesses, because we do not know what specific countries will participate in the water polo events.

We also were not given a time interval to start for the different Olympic sports so our schedule would have to be adjusted to that based on the other sports.

8. Concluding Paragraph

The priority of this project was limiting the number of medals and flags while also having enough so that the IOC is prepared for all circumstances. A schedule was constructed with 5-minute blocks (some are event + small break) to ensure that the schedule can be easily read, followed, and implemented. We then used this schedule to help decide how many flags and medals we needed. We also collected data from past Olympic games to aid with predictions. It was decided that every country that has placed on the podium in the past 4 games will get 3 flags, and every country that has placed 4th will get 1 flag. This way, if a country places multiple times on the podium, the IOC is prepared. If a 4th place winner from previous years comes back to the 2024 Games stronger than ever before, the IOC is also prepared. The total comes out to 105 flags for 39 countries. When calculating medals, the number of medals and ties were counted over the past 7 Olympic games. Standard deviation was then applied to determine the number of ties that can happen. This ensures a 95% certainty on how the results might turn out. Whenever necessary, the calculation was rounded up to 1 in case of the worst-case scenario and there is indeed a tie. Because it's unclear which tie it will be (tie for gold, tie for silver, or tie for bronze), extra medals of each type will be prepared. The total comes out to be 26 sets of medals for 16 events. All of our data was organized into data tables to make it easy to read and analyze.



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9. Letter

Dear IOC,

Thank you for choosing to work with the MidMCM Olympic Consultants. After collecting much data and performing many calculations, we have found a solution.

For the schedule, our team has chosen the La Défense Arena (Swimming and Water Polo). We assembled an optimized model and schedule, accounting for as many outcomes as possible to ensure smooth ceremonies and avoid mishaps and missing medals/flags, while also keeping a budget in mind. In order to do this, we began with creating a schedule for the ceremonies. Each swimming event will take 2-20 minutes depending on the type of event (2 minutes for 100m butterfly, 3 minutes for 200m butterfly, 10 minutes for 800m freestyle, 20 min for 1500m freestyle, etc.). There will be small breaks between events to ensure that each athlete gets ample rest before competing again. Each of these breaks will be fitted in so that the entire schedule can be easily organized into 5-minute blocks. This kind of schedule is not only easy to follow, but it is also flexible, allowing for minor modifications if needed.

We then assembled and analyzed data and trends in placings from past Summer Olympic games for our chosen events, in order to help decide how many medals and flags are needed. There will be 3 flags for every country that placed on the podium in the last 4 summer Olympic games (Australia, Austria, Belarus, Belgium, Brazil, Canada, China, Denmark, Finland, France, Germany, Great Britain, Hong Kong, Hungary, Italy, Japan, Kazakhstan, Lithuania, Netherlands, Norway, ROC, Serbia, Singapore, Slovenia, South Africa, South Korea, Spain, Sweden, Switzerland, Tunisia, Ukraine, USA, and Zimbabwe), and 1 for every country that has placed 4th because they still have a chance of placing. (Croatia, Greece, Montenegro, New Zealand, Poland, and Romania). Our final total comes out to 105 flags. When predicting the required medals, it was important that we not only consider all events, but also account for ties on the podium. To do this, we collected data from the past 7 summer Olympic games and took notes of the podium placings and counted ties. We then applied a fundamental concept of statistics and probability, standard deviation, to predict the number of ties. This final total comes out to be 26 sets of medals (gold, silver, bronze) across 16 events.

If similar processes (data collection and calculations) were conducted on events for the other 2 arenas, it should produce a functioning result.

Thank you again for working with the MidMCM Olympic Consultants. We hope this will aid you in securing smooth sailing for the 2024 Paris summer Olympics. It was a pleasure to collaborate with you.

Sincerely,
Team 14255



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10. List of References

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